BJT COMPLETE STUDY GUIDE

Chapter 3: Bipolar Junction Transistors (Sections 3.1–3.7)
Chapter 4: DC Biasing — BJTs (Sections 4.1–4.5)

Exam-focused | Concept-first | Fully worked problems

CHAPTER 3: BIPOLAR JUNCTION TRANSISTORS

3.1 & 3.2 Introduction and Transistor Construction

The **Bipolar Junction Transistor (BJT)** was invented on December 23, 1947 by Shockley, Brattain, and Bardeen at Bell Labs. It is a **three-layer, three-terminal semiconductor device** that forms the foundation of analog amplification and switching circuits. The word *bipolar* reflects that both holes AND electrons participate in current flow.

TWO TYPES OF BJTS

npn Transistor	pnp Transistor
Structure: n-p-n layers	Structure: p-n-p layers
Majority carriers: electrons	Majority carriers: holes
Arrow in symbol points OUT of base (Not Pointing In)	Arrow in symbol points IN to base (Pointing In)
More commonly used in practice	Less common; all polarities reversed vs. npn

EXAM TIP

To remember arrow direction: npn = "Not Pointing In"; pnp = "Pointing In".

Key structural fact: The base layer is extremely thin (0.001 in.) compared to the outer layers (0.150 in.), giving a ratio of 150:1. The base is also lightly doped, which drastically limits current flow through it.

TERMINAL NAMES

Terminal	Role
E = Emitter	Heavily doped; source/sink of majority carriers; always largest current
B = Base	Very thin, lightly doped; controls transistor action; always smallest current
C = Collector	Collects majority carriers that cross the base; nearly equal to I_E

3.3 Transistor Operation

For normal (active) transistor operation, the two p-n junctions must be biased differently: the **base-emitter junction is forward-biased** and the **base-collector junction is reverse-biased**.

Core Operating Principle

When BE is forward-biased, majority carriers flood from emitter into the thin base. Because the base is so thin and lightly doped, very few recombine there. The vast majority (>95%) are swept across the reverse-biased BC junction into the collector. This is WHY $I_C \approx I_E$.

CURRENT FLOW (KCL AT THE TRANSISTOR NODE)

KCL	$I_E = I_C + I_B$	*Always true*

Relative magnitudes: I_E is largest (mA range), I_B is smallest (μA range, ~1% of I_E), $I_C \approx I_E$ (milliamperes).

Collector current has two components: $I_C = I_C(\text{majority}) + I_{CO}(\text{minority})$. The minority component I_{CO} is the leakage current (also called I_{CBO}). It is temperature-sensitive and generally negligible in modern transistors.

3.4 Common-Base (CB) Configuration

In common-base, the **base is shared** between input (emitter side) and output (collector side). Input is I_E, output is I_C.

ALPHA (A) — THE CB CURRENT GAIN

DC Alpha	$\alpha_{dc} = I_C / I_E$	*Always < 1 (typically 0.90–0.998)*
With leakage	$I_C = \alpha \cdot I_E + I_{CBO}$	*Full collector current equation*
AC Alpha	$\alpha_{ac} = \Delta I_C / \Delta I_E \quad \vert \quad V_{CB} = \text{const}$	*For AC analysis*

Why α < 1?

Not all emitter carriers make it to the collector — a tiny fraction recombine in the base, forming I_B. So I_C is always slightly less than I_E. Typical α = 0.990 to 0.998.

CB CHARACTERISTIC REGIONS (OUTPUT: I_C VS. V_{CB} AT CONSTANT I_E)

Region	Conditions	Key Behavior
Active	BE: forward-biased | BC: reverse-biased	$I_C \approx I_E$; linear amplification region; I_C nearly independent of V_{CB}

Saturation	$V_{CB} < 0$ V (both junctions forward-biased)	I_C rises sharply; left side of graph
Cutoff	$I_E = 0$ mA (BE reverse-biased)	$I_C = I_{CBO} \approx 0$ mA; transistor OFF
Breakdown	$V_{CB} > BV_{CBO}$	Avalanche effect; large current surge; device can be destroyed

IMPORTANT APPROXIMATION FOR DC ANALYSIS

Active Region	$I_C \approx I_E$ and $V_{BE} \approx 0.7$ V	*Used in all DC analysis*

EXAM TIP

Once a transistor is confirmed to be in the active (ON) state, immediately write $V_{BE} = 0.7$ V. This is the single most important shortcut in BJT analysis.

3.5 Common-Emitter (CE) Configuration — Most Important!

In common-emitter, the **emitter is shared** between input (base side) and output (collector side). This is by far the most widely used configuration because it provides significant **voltage, current, and power amplification**.

BETA (B) — THE CE CURRENT GAIN

DC Beta (hFE)	$\beta_{dc} = I_C / I_B$	*Typical range: 50 to 400+*
AC Beta (hfe)	$\beta_{ac} = \Delta I_C/\Delta I_B$ | $V_{CE} = \text{const}$	*Used for AC small-signal*

DERIVED CURRENT EQUATIONS FROM B

Collector current	$I_C = \beta \cdot I_B$	*Direct scaling*
Emitter current	$I_E = (\beta + 1) \cdot I_B$	*Follows from KCL*

RELATING A AND B — CRITICAL FORMULAS!

Formula	Derivation / Notes
$\alpha = \beta / (\beta + 1)$	Derived from $\alpha = I_C/I_E$ and $I_E = (\beta+1)I_B$
$\beta = \alpha / (1 - \alpha)$	Algebraic rearrangement
$I_{CEO} = (\beta+1) \cdot I_{CBO} \approx \beta \cdot I_{CBO}$	Leakage in CE config is β times larger than CB leakage

Why I_{CEO} Is Larger Than I_{CBO}

In common-emitter, leakage current I_{CBO} (from the BC junction) gets multiplied by β as it passes through the transistor, giving $I_{CEO} = \beta \cdot I_{CBO}$. This is why CE cutoff is defined as $I_C = I_{CEO}$, not zero.

CE CHARACTERISTIC REGIONS (OUTPUT: I_C VS. V_{CE} AT CONSTANT I_B)

Region	Junction Conditions	Notes
Active	BE: forward | BC: reverse	$I_C = \beta \cdot I_B$; linear operation; I_B curves are nearly flat
Saturation	BE: forward | BC: forward	$V_{CE} < V_{CE(sat)} \approx 0.3$ V; I_C at maximum; signal distortion
Cutoff	$I_B = 0$ μA → $I_C = I_{CEO}$	Both junctions reverse-biased; transistor is OFF
Breakdown	$V_{CE} > BV_{CEO}$	Avalanche + punch-through; $BV_{CEO} \approx ½ \cdot BV_{CBO}$

EXAM TIP

The active region biasing conditions are the SAME for both CB and CE: BE forward-biased, BC reverse-biased. Only the characteristic curves (what you plot) differ.

Determining β_{ac} from CE Characteristic Curves

Choose a Q-point. Draw a vertical line at that V_{CE}. Pick I_B curves symmetrically above and below. Measure ΔI_C and ΔI_B.

β_{ac} calculation	$\beta_{ac} = \Delta I_C / \Delta I_B$ (at const. V_{CE})

β_{ac} and β_{dc} are usually approximately equal (treated as interchangeable in analysis). Exact equality holds when $I_{CEO} = 0$.

3.6 Common-Collector (CC) Configuration

The **collector is tied to ground** (for AC). Input enters at the base, output is taken from the emitter. Also called the **emitter-follower**.

Property	Value / Description
Input impedance	HIGH (like CE)
Output impedance	LOW (opposite of CB and CE)
Voltage gain	≈ 1 (output follows input — hence "emitter follower")
Current gain	β + 1 (high)
Primary use	Impedance matching between stages

Analysis approach	Use CE characteristics; output is I_E vs. V_{CE}

3.7 Limits of Operation

Every BJT has a safe operating region defined by three boundaries. Operation outside these limits causes distortion or device destruction.

Limit Parameter	Definition & Importance
I_{Cmax} (max collector current)	Horizontal line on characteristics. Exceeding this can physically damage the device.
V_{CEmax} (= BV_{CEO})	Vertical line on characteristics. Exceeding this triggers avalanche breakdown.
$P_{Cmax} = V_{CE} \times I_C$	A hyperbolic curve on the I_C-V_{CE} graph. The product must stay below this at ALL operating points.
$V_{CE(sat)} \approx 0.3$ V	Left boundary. Below this, transistor enters saturation and signal is distorted.
$I_C = I_{CEO}$ (cutoff)	Lower boundary for linear operation.

Max power	$\mathbf{P_{Cmax} = V_{CE} \times I_C}$	***Must not be exceeded***

To plot the power curve: Pick three V_{CE} values (including V_{CEmax} and ½·I_{Cmax} point), calculate $I_C = P_{Cmax} / V_{CE}$, and connect the points. This hyperbola is the upper-right boundary of safe operation.

Safe operating region requires ALL THREE simultaneously:

$I_{CEO} \le I_C \le I_{Cmax}$
$V_{CE(sat)} \le V_{CE} \le V_{CEmax}$
$V_{CE} \times I_C \le P_{Cmax}$

Chapter 3 — Complete Formula Reference

Formula	Equation	When to Use
KCL	$I_E = I_C + I_B$	Always
V_{BE} (active)	$V_{BE} \approx 0.7$ V	Any active-region DC analysis
I_C approximation	$I_C \approx I_E$	When I_B negligible
DC Alpha	$\alpha_{dc} = I_C / I_E$	Common-base analysis
Full I_C (CB)	$I_C = \alpha \cdot I_E + I_{CBO}$	Includes leakage
DC Beta	$\beta_{dc} = I_C / I_B$	Common-emitter analysis
From beta	$I_C = \beta \cdot I_B$	Find I_C from I_B

From beta	$I_E = (\beta+1)\cdot I_B$	Find I_E from I_B
Alpha from beta	$\alpha = \beta/(\beta+1)$	Convert between parameters
Beta from alpha	$\beta = \alpha/(1-\alpha)$	Convert between parameters
CE leakage	$I_{CEO} \approx \beta\cdot I_{CBO}$	Cutoff condition in CE
Max power	$P_{Cmax} = V_{CE}\times I_C$	Stay below this curve
CB power limit	$P_{Cmax} = V_{CB}\times I_C$	CB saturation

Chapter 3 — Selected Key Problems with Full Solutions

PROBLEM 3-A

Given: I_E = 8 mA, and I_B = (1/100) × I_C. Find I_C and I_B.

Why Important: Tests KCL application, the relationship between I_B and I_C, and solving a system of equations — foundational for all transistor analysis.

SOLUTION

Step 1: Set up the given relationship: $I_B = I_C / 100$

Step 2: Apply KCL: $I_E = I_C + I_B \rightarrow 8\text{ mA} = I_C + I_C/100 = I_C(1 + 0.01) = 1.01\cdot I_C$

Step 3: Solve: $I_C = 8\text{ mA} / 1.01 = 7.92\text{ mA}$

Step 4: Find I_B: $I_B = I_C / 100 = 7.92\text{ mA} / 100 = 0.0792\text{ mA} = 79.2\ \mu\text{A}$

Step 5: Verify: $I_E = 7.92 + 0.0792 \approx 8\text{ mA}$

PROBLEM 3-B

Given α_{dc} = 0.998, find I_C if I_E = 4 mA. Also: find α_{dc} if I_E = 2.8 mA, I_C = 2.75 mA, I_{CBO} = 0.1 μA.

Why Important: Direct application of α_{dc}. Also reinforces that I_{CBO} is tiny and usually ignored. Tests both directions of the alpha formula.

SOLUTION — PART (A)

Step 1: Use $I_C = \alpha\cdot I_E + I_{CBO} \approx \alpha\cdot I_E$ (ignoring tiny I_{CBO})

Step 2: $I_C = 0.998 \times 4\text{ mA} = 3.992\text{ mA} \approx 3.99\text{ mA}$

SOLUTION — PART (B)

Step 1: Use full formula: $I_C = \alpha\cdot I_E + I_{CBO}$

Step 2: $2.75\text{ mA} = \alpha \times 2.8\text{ mA} + 0.1\times10^{-3}\text{ mA}$

Step 3: $\alpha\cdot 2.8\text{ mA} = 2.75 - 0.0001 = 2.7499\text{ mA}$

Step 4: $\alpha = 2.7499 / 2.8 = 0.9821$

PROBLEM 3-C

On CE characteristics with Q-point at $I_B = 25\ \mu A$, $V_{CE} = 7.5$ V: the $I_B = 20\ \mu A$ curve gives $I_C \approx 2.2$ mA, $I_B = 30\ \mu A$ gives $I_C \approx 3.2$ mA, and the Q-point $I_C \approx 2.7$ mA. Find β_{ac} and β_{dc}.

Why Important: This is the fundamental problem for reading CE characteristics. Tests β_{ac} (change-based) vs β_{dc} (ratio-based) and shows why they differ — a very commonly tested concept.

SOLUTION

Step 1: For β_{ac}: choose two symmetric I_B values around Q-point at constant $V_{CE} = 7.5$ V

Step 2: $\Delta I_C = I_{C2} - I_{C1} = 3.2\ mA - 2.2\ mA = 1.0\ mA$

Step 3: $\Delta I_B = I_{B2} - I_{B1} = 30\ \mu A - 20\ \mu A = 10\ \mu A$

Step 4: $\beta_{ac} = \Delta I_C / \Delta I_B = 1.0\ mA / 10\ \mu A = 100$

Step 5: For β_{dc}: use Q-point values directly

Step 6: $\beta_{dc} = I_{CQ} / I_{BQ} = 2.7\ mA / 25\ \mu A = 108$

Step 7: Conclusion: $\beta_{ac} \approx \beta_{dc}$ (100 ≈ 108); they're close but not equal — perfectly normal.

PROBLEM 3-D

(a) Find β_{dc} if $\alpha_{dc} = 0.980$. (b) Find α if $\beta = 120$. (c) If $\beta = 120$ and $I_C = 2.0$ mA, find I_E and I_B.

Why Important: Tests the critical α–β conversion formulas, which appear in almost every BJT problem.

SOLUTION

Step 1: (a) $\beta = \alpha/(1-\alpha) = 0.980 / (1 - 0.980) = 0.980 / 0.020 = 49$

Step 2: (b) $\alpha = \beta/(\beta+1) = 120 / (120+1) = 120/121 = 0.9917$

Step 3: (c) From $I_C = \beta \cdot I_B \rightarrow I_B = I_C/\beta = 2.0\ mA / 120 = 16.67\ \mu A$

Step 4: $I_E = (\beta+1) \cdot I_B = 121 \times 16.67\ \mu A = 2.017\ mA$

Step 5: Verify: $I_E = I_C + I_B = 2.0 + 0.01667 = 2.017$ mA

PROBLEM 3-E

A transistor has $I_{Cmax} = 6$ mA, $BV_{CEO} = 15$ V, $P_{Cmax} = 35$ mW. Sketch the safe operating region on the CE characteristics (similar to Fig. 3.13).

Why Important: Tests understanding of all three operating limits simultaneously and the shape of the power dissipation hyperbola.

SOLUTION

Step 1: Horizontal limit: I_C = 6 mA (draw horizontal line)

Step 2: Vertical limit: V_{CE} = 15 V (draw vertical line)

Step 3: Power curve: $V_{CE} \times I_C$ = 35 mW. Find three points:

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At I_C = 6 mA: V_CE = 35/6 = 5.83 V → point (5.83 V, 6 mA)
At V_CE = 15 V: I_C = 35/15 = 2.33 mA → point (15 V, 2.33 mA)
At I_C = 3 mA: V_CE = 35/3 = 11.67 V → point (11.67 V, 3 mA)
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Step 4: The safe region is below all three boundaries simultaneously (upper-left region bounded by the hyperbola, horizontal line at 6 mA, and vertical line at 15 V).

CHAPTER 4: DC BIASING — BJTs (Sections 4.1–4.5)

4.1 Introduction — Why DC Biasing?

The Fundamental Principle

Any increase in AC voltage, current, or power from a transistor amplifier is NOT magic — it is a transfer of energy from the DC power supply. The DC bias sets the operating point; the AC signal rides on top of it.

The three foundational equations used in **all** DC bias analysis:

(4.1) V_{BE}	$V_{BE} \approx 0.7$ V	*Active region assumption*
(4.2) I_E	$I_E = (\beta+1) \cdot I_B \approx I_C$	*KCL rearranged*
(4.3) I_C	$I_C = \beta \cdot I_B$	*Beta relationship*

Procedure for ALL bias configurations: First find I_B (using KVL in the base-emitter loop). Then use $I_C = \beta \cdot I_B$ to find I_C. Then use KVL in the collector-emitter loop to find V_{CE}.

4.2 Operating Point (Q-Point)

The **Q-point (Quiescent Point)** is the DC operating point — the steady-state I_C and V_{CE} when no AC signal is applied. It must be placed in the **active region** for linear amplification.

Q-Point Location	Effect
Point A (origin): no bias	Device is OFF; can only amplify positive half of signal
Point B (center of active region)	BEST: maximum symmetrical output swing; most linear operation
Point C (near saturation)	Limited positive swing; non-linear β in this region
Point D (near V_{CEmax})	Limited positive voltage swing; risk of breakdown

BIASING CONDITIONS SUMMARY

Region	BE Junction	BC Junction
Active (linear amplification)	Forward-biased (+0.7 V)	Reverse-biased
Saturation	Forward-biased	Forward-biased
Cutoff	Reverse-biased	Reverse-biased

EXAM TIP

Temperature sensitivity: Higher temperature → larger I_{CEO} → Q-point shifts. A good bias circuit must be stable against both β variation and temperature changes.

4.3 Fixed-Bias Configuration

Circuit: R_B connected from V_{CC} to base; R_C connected from V_{CC} to collector. Simplest configuration but Q-point is highly sensitive to β variations.

STEP-BY-STEP ANALYSIS

Step	Equation & Calculation
1. Apply KVL to Base-Emitter loop	$V_{CC} - I_B R_B - V_{BE} = 0 \rightarrow I_B = (V_{CC} - V_{BE})/R_B$
2. Find I_C	$I_C = \beta \cdot I_B$
3. Apply KVL to Collector-Emitter loop	$V_{CE} = V_{CC} - I_C R_C$
4. Find V_B, V_C, V_E	$V_B = V_{BE} = 0.7$ V; $V_C = V_{CE}$ (since $V_E = 0$); $V_E = 0$ V
5. Check: BC junction reversed?	$V_{BC} = V_B - V_C$ should be negative (confirming active region)

FIXED-BIAS KEY FORMULAS

Base current (4.4)	$I_B = (V_{CC} - V_{BE}) / R_B$	*$V_{BE} = 0.7$ V*
Collector current (4.5)	$I_C = \beta \cdot I_B$	
C-E voltage (4.6)	$V_{CE} = V_{CC} - I_C R_C$	
Saturation current (4.11)	$I_{Csat} = V_{CC} / R_C$	*Set $V_{CE} = 0$ V; max I_C*

Fixed-Bias Weakness

$I_B = (V_{CC} - V_{BE}) / R_B$ is FIXED by R_B and V_{CC}. If β changes (e.g., temperature increase), $I_C = \beta \cdot I_B$ changes proportionally. This causes the Q-point to shift dramatically — poor stability.

LOAD-LINE ANALYSIS FOR FIXED-BIAS

The load line is a straight line on the I_C-V_{CE} graph connecting two points:

Point	Calculation
I_C-axis intercept (V_{CE} = 0 V)	$I_C = V_{CC} / R_C$ (this is also I_{Csat})
V_{CE}-axis intercept (I_C = 0 mA)	$V_{CE} = V_{CC}$

The intersection of this load line with the I_B curve determined by the bias circuit defines the Q-point (I_{CQ}, V_{CEQ}).

EXAM TIP

Load-line analysis works for ANY configuration. The two intercepts are always at $I_C = V_{CC}/(R_C+R_E)$ and $V_{CE} = V_{CC}$. Changing R_B moves the Q-point along the fixed load line.

4.4 Emitter-Bias Configuration

Circuit: Same as fixed-bias but with an emitter resistor R_E added. This is the key improvement. R_E provides negative feedback that stabilizes the Q-point against β variations.

Base-Emitter loop analysis: KVL around base-emitter loop: $V_{CC} - I_BR_B - V_{BE} - I_ER_E = 0$

Substituting $I_E = (\beta+1)\cdot I_B$ and solving for I_B:

I_B formula (4.17)	$I_B = (V_{CC} - V_{BE}) / [R_B + (\beta+1)\cdot R_E]$	*Note: R_E is "reflected" by (β+1)*

The Reflected Resistance Concept

R_E in the emitter appears as $(\beta+1)\cdot R_E$ in the base-emitter loop. This is because $I_E = (\beta+1)\cdot I_B$ passes through R_E, but we're solving for I_B. Memorize: $R_i = (\beta+1)\cdot R_E$. This concept recurs in all emitter-resistor configurations.

COLLECTOR-EMITTER LOOP & NODE VOLTAGES

V_{CE} (4.19)	$V_{CE} = V_{CC} - I_C(R_C + R_E)$	*Both R_C and R_E in loop*

Voltage	Formula
V_E (emitter to ground)	$V_E = I_ER_E \approx I_CR_E$
V_C (collector to ground)	$V_C = V_{CC} - I_CR_C$ or $V_C = V_{CE} + V_E$
V_B (base to ground)	$V_B = V_{BE} + V_E$ or $V_B = V_{CC} - I_BR_B$
V_{BC} (check: must be negative)	$V_{BC} = V_B - V_C$ (negative = active region)

Saturation current (4.25)	$I_{Csat} = V_{CC} / (R_C + R_E)$

Why is emitter-bias more stable? If β increases: I_C tries to increase → $V_E = I_C R_E$ increases → V_B is fixed (determined by R_B) → $V_{BE} = V_B - V_E$ decreases → I_B decreases → I_C is pulled back down. This self-correcting feedback is called emitter degeneration.

4.5 Voltage-Divider Bias Configuration — Most Stable!

Circuit: Two resistors R_1 and R_2 form a voltage divider from V_{CC} to ground, setting the base voltage. R_E is still present. This is the most stable and most commonly used bias circuit in practice.

Method 1: Exact Analysis (Thévenin Equivalent)

Reduce the voltage divider to a Thévenin equivalent looking into the base:

R_{Th} (4.28)	$R_{Th} = R_1 \parallel R_2 = (R_1R_2)/(R_1+R_2)$	*V_{CC} shorted = 0 V*
E_{Th} (4.29)	$E_{Th} = V_{CC} \cdot R_2/(R_1+R_2)$	*Voltage divider rule*

Then the equivalent circuit is identical to emitter-bias with $V_{CC} \to E_{Th}$ and $R_B \to R_{Th}$:

I_B (4.30)	$I_B = (E_{Th} - V_{BE}) / [R_{Th} + (\beta+1) \cdot R_E]$	*Same form as (4.17)!*
V_{CE} (4.31)	$V_{CE} = V_{CC} - I_C(R_C + R_E)$	*Same as emitter-bias*

Method 2: Approximate Analysis (Faster — use when condition is met)

Condition for Approximate Method

Approximate analysis is valid when: $\beta \cdot R_E \geq 10 \cdot R_2$. This ensures $I_B \ll I_2$, so the base draws negligible current from the voltage divider. When satisfied, the base voltage is set almost entirely by R_1 and R_2.

Condition (4.33)	$\beta \cdot R_E \geq 10 \cdot R_2$	*Check this FIRST!*

If condition is satisfied, use this fast sequence:

Step	Formula
1. Base voltage (voltage divider only)	$V_B = V_{CC} \cdot R_2/(R_1+R_2)$
2. Emitter voltage	$V_E = V_B - V_{BE}$ $(= V_B - 0.7\ V)$

3. Emitter current (≈ collector current)	$I_{CQ} \approx I_E = V_E/R_E$
4. Collector-emitter voltage	$V_{CEQ} = V_{CC} - I_C(R_C + R_E)$
Note	β does NOT appear! Q-point is nearly β-independent — this is the key advantage.

Power of the approximate method: Because β is eliminated from the equations, the Q-point is essentially independent of β. Even if β changes by 50% (from 100 to 50), I_{CQ} changes by less than 3%!

> **EXAM TIP**
>
> Always check $\beta \cdot R_E \geq 10 \cdot R_2$ before using the approximate method. If not satisfied, use the exact Thévenin method. Both give $V_{CE} = V_{CC} - I_C(R_C+R_E)$.

STABILITY COMPARISON — ALL FOUR CONFIGURATIONS

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